

Chapter 5

Computer Modeling of MOSFETs

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I. INTRODUCTION

Computer modeling of MOSFETs has been an active field during the 1970s. This activity has intensified since the advent of very large scale integration (VLSI) of MOSFET circuits with the goal of producing chips containing hundreds of thousands of devices whose feature size may be $1\text{ }\mu\text{m}$ or less. Experimental investigation of problems involving such a large

number of devices would be overly time-consuming, expensive, and in certain aspects not feasible. Today, computer-aided design (CAD) has become an essential tool in VLSI endeavors. It can be divided into three areas: process, device, and circuit, all of which are essential in VLSI technology.

Process models [1–5] simulate the formation of the device. They delineate the composition and physical dimensions, such as doping profiles and oxide thickness, when a wafer is processed. Circuit models [6,7] simulate time-dependent action of interconnected devices. Thus, they predict the responses of elementary portions of large circuits. Between process and circuit models are device models, which can be analytical or numerical. Analytical models [8–16] are often used in conjunction with circuit models. Because of the nonlinearity and multidimensional nature of the very small devices associated with VLSI, simplifying assumptions are necessary to render these models analytically tractable. Parameters are introduced to account for the essential physical effects of the device. Although useful as elements in circuit modeling and explaining device behavior, present analytical models are limited in their ability to quantitatively predict device performance.

In device modeling, which is the topic of this chapter, one solves the Poisson and continuity equations with boundary conditions appropriate to the device. Within the context of numerical techniques, these equations can be solved exactly, accounting for nonlinear effects and multidimensionality. Most MOSFET computer models are static and two-dimensional [17–28], although transient [29], three-dimensional [30] models have also been reported. Strictly speaking, very small MOSFETs are essentially three-dimensional, requiring the treatment of at least short-channel and narrow-width effects [30–32]. However, at present, the three-dimensional model is not practical in CAD because of computer limitations. A practical compromise between utility and realistic simulation is the two-dimensional static model. It can be shown that for devices with gate width one order of magnitude larger than the gate length, a two-dimensional model can predict device performance accurately. Thus optimization of design and diagnostics can be done with confidence.

Although many two-dimensional computer MOSFET models exist, they vary in numerical stability, accuracy, and computational efficiency. Recently published models, which tend to be more reliable, do not give details of formulation or numerical techniques. It is our objective to present a bulk Si MOSFET model in comprehensive detail. The model is capable of simulating MOSFETs for VLSI applications with high accuracy under normal operating conditions. Given the device physical parameters, the external characteristics as well as internal electrostatic

potential and carrier distributions can be predicted. Thus, the model can be used for design and diagnostic purposes. Additional physics, such as hot electron effects [33,34], carrier generation [21] and recombination [35], and SOS-MOSFET can be conveniently incorporated. Device parameters can be extracted from modeling results as desired [36,37]. Some outstanding features of our model as compared with others are the grid system, the computational algorithm, and the treatment of the Si-oxide interface.

The partial differential equations governing the device physics in a MOSFET are discretized using finite difference techniques on a nonuniform grid. The grid system is designed to produce high resolution in the channel region. A very small grid spacing (approximately $0.01\ \mu\text{m}$) is used near the oxide-semiconductor interface. To account for several micrometers of substrate while keeping within a manageable number of grid points, the grid size is increased in a geometrical progression in the direction perpendicular to the oxide-semiconductor interface (x direction). Uniform grid size is employed in the direction parallel to the interface (y direction) for accurate determination of surface potential along the channel. This is in contrast with most existing models, where fine resolutions are used in the drain-substrate and source-substrate junctions, while coarse grids are used in the channel.

As the grid size increases in the x direction, the aspect ratio of the rectangular grid changes. It is important to maintain the aspect ratio $\Delta x/\Delta y$ near unity, otherwise the discretized Poisson's equation loses its two-dimensional property. To enforce this condition, the grid size in the y direction is doubled whenever Δx becomes too large compared to Δy (see Fig. 4), resulting in grid-line terminations. At these termination points, Poisson and continuity equations are represented by six-point and seven-point formulas, respectively. This deviation from the standard five-point formula prevents the use of Stone's SIP technique [38] for fast matrix inversion, a price we pay for accuracy. However, SOR, especially SLOR together with a modified Gummel algorithm, work quite satisfactorily, as demonstrated by others [35,39].

The Gummel algorithm [40,41] is extended to two dimensions for computation of Poisson and continuity equations. Our technique differs from that of Slotboom [35] in that the Poisson grid and the continuity grid are interspersed but equal in size. We have devised a new technique to effect finer resolution of electrostatic potential where required, such as in the junction regions. This new technique, discussed in Section II, permits spatial resolution approaching the Debye length. However, the main Poisson matrix, except for the oxide regions, remains the same size as the continuity matrix, thus greatly improving computational efficiency.

The oxide–semiconductor interface has been a problem in MOSFET modeling since it is governed neither by Laplace's nor by Poisson's equation. A potential equation was derived specifically for the interface, allowing a unified treatment for the whole device.

In the sections to follow, the models for MOSFET and computation algorithm are presented. In the computation, the source instead of the substrate is chosen as the reference of the electrostatic potential so that the model can be easily extended to SOS MOSFET, in which the substrate is floating [42,43]. The physical validity and accuracy are demonstrated by comparing modeled and experimental results. In addition, the model exhibits well-known MOSFET device characteristics. Short-channel effects, which are an important consideration in VLSI, are examined and their physical origins explained. Finally, parasitic resistance [44] effects are modeled, which indicate that the Si MOSFET, beside other considerations [25,45], may not be practical with channel length less than $0.5\text{ }\mu\text{m}$ and that the constant field scaling rules [45] cannot predict submicrometer device performance because of parasitic effects.

II. MOSFET DEVICE MODELING

A. Mathematical Model

The state of a static two-dimensional MOSFET can be described by the Poisson and continuity equations as

$$\begin{aligned} \nabla^2\psi &= (q/\epsilon)(p - p^0 - n + n^0), \\ \nabla \cdot J^n &= -q(G^n - R^n), \quad \nabla \cdot J^p = q(G^p - R^p), \end{aligned} \quad (1)$$

where ψ is the electrostatic potential, p and n the free hole and free electron densities, p^0 and n^0 the acceptor and donor doping densities, q and ϵ the magnitude of the electronic charge and the dielectric constant, G and R the generation and recombination rates, and J^n and J^p the electron and hole current densities. The current densities are given by

$$\begin{aligned} J^n &= -q\mu^n n \nabla\psi + qD^n \nabla n, \\ J^p &= -q\mu^p p \nabla\psi - qD^p \nabla p, \end{aligned}$$

where the mobility μ and the diffusion coefficient D are related by the Einstein relation $D = (kT/q)\mu$, and kT is the thermal energy.

The state equations as presented describe the classical carrier transport process in semiconductor devices; thermal equilibrium is assumed and Boltzmann statistics prevail. Deviations from these conditions, such as velocity saturation, are accounted for by introducing the carrier mobilities

as a function of the electrostatic field, doping densities, etc. Consequently, this approach is sometimes referred to as the *mobility model* [23]. The model is justified on the basis of agreement between computational and experimental results. For very small devices, where ballistic carrier transport becomes significant, the validity of the mobility model may be questionable. This size limitation may occur below $\sim 0.25 \mu\text{m}$ for Si and $\sim 0.5 \mu\text{m}$ for GaAs devices. For these small dimensions, a Monte Carlo-type model would be more appropriate [47,48].

For computational purposes, it is more convenient to introduce a system of units in which the voltage is measured in kT/q , the carrier density in the units of the intrinsic carrier density n_i , and the distance in $(\epsilon kT/q^2 n_i)^{1/2}$. Using these units and incorporating the Einstein relation, the state equations become

$$\begin{aligned} \nabla^2 \psi &= n - n^0 - p + p^0, \\ \nabla \cdot J^n &= -(G^n - R^n), \quad \nabla \cdot J^p = G^p - R^p, \\ J^n &= -(\alpha^n)^{-1}(n \nabla \psi - \nabla n), \quad J^p = -(\alpha^p)^{-1}(p \nabla \psi - \nabla p), \end{aligned} \quad (2)$$

where α^{-1} is the normalized carrier mobility.

The state equations (2) together with the boundary conditions, doping distributions, generation and recombination rates, and functional dependence of α^{-1} determine the fundamental quantities ψ , n , and p in a semiconductor device. All other quantities of interest can be derived from these fundamental quantities.

It is sometimes more convenient for purposes of algebraic manipulation to use the quasi-Fermi potentials ϕ^n and ϕ^p to describe the electron and hole densities as

$$n = \exp(\psi - \phi^n), \quad p = \exp(\phi^p - \psi).$$

Following Slotboom [35], Φ^n and Φ^p are defined as

$$\begin{aligned} \Phi^n &= \exp(-\phi^n) \quad \text{such that} \quad n = \Phi^n e^\psi, \\ \Phi^p &= \exp(\phi^p) \quad \text{such that} \quad p = \Phi^p e^{-\psi}. \end{aligned} \quad (3)$$

With the new notations, the state equations become

$$\begin{aligned} \nabla^2 \psi &= \Phi^n e^\psi - \Phi^p e^{-\psi} - n^0 + p^0, \\ \nabla \cdot J^n &= -(G^n - R^n), \quad \nabla \cdot J^p = G^p - R^p, \\ J^n &= (\alpha^n)^{-1} e^\psi \nabla \Phi^n, \quad J^p = -(\alpha^p)^{-1} e^{-\psi} \nabla \Phi^p. \end{aligned} \quad (4)$$

Thus the fundamental quantities are ψ , Φ^n , and Φ^p , compared with the set used in (2), ψ , n , and p . The two sets are, of course, equivalent and can be used interchangeably for computational convenience. Note, however, that Φ^n and Φ^p are not quasi-Fermi potentials per se but are alternate representations defined by Eq. (3).

The state equations thus describe the bipolar carrier transport, generation, and recombination, which are believed necessary in modeling devices such as SOS MOSFET [42,44]. For bulk Si devices, the R and G terms can be neglected so that, for an n -channel device,

$$\begin{aligned}\nabla^2\psi &= \Phi^n e^\psi - \Phi^p e^{-\psi} - n^0 + p^0, \\ \nabla \cdot J^n &= 0, \\ J^n &= (\alpha^n)^{-1} e^\psi \nabla \Phi^n, \\ \nabla \Phi^p &= 0,\end{aligned}\tag{5}$$

and for a p -channel device,

$$\begin{aligned}\nabla^2\psi &= \Phi^n e^\psi - \Phi^p e^{-\psi} - n^0 + p^0, \\ \nabla \cdot J^p &= 0, \\ J^p &= -(\alpha^p)^{-1} e^{-\psi} \nabla \Phi^p, \\ \nabla \Phi^n &= 0.\end{aligned}\tag{6}$$

Henceforth, only n -channel devices will be discussed; p -channel devices can be similarly treated. The boundary conditions and doping profiles appropriate for an n -channel device are schematically shown in Fig. 1. The source, instead of substrate, potential is used as the reference for convenience in extending the present model to SOS MOSFET, which has

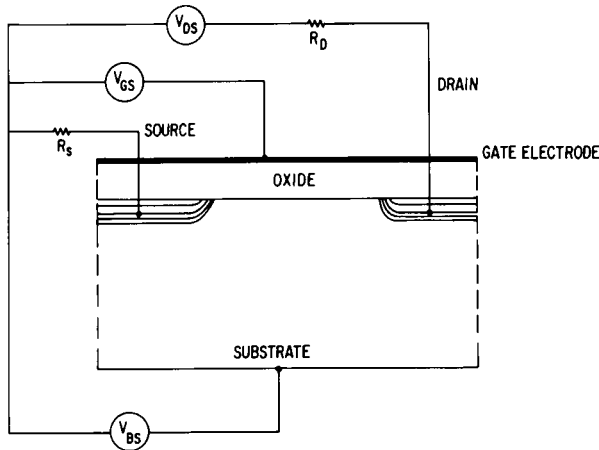


Fig. 1. Schematic representation of the n -channel MOSFET device featuring the n^+ doping contours for the source and drain. The source and drain boundaries are picked at an n^+ contour level; any n^+ doping level higher than this is regarded as metallic. These internal "metallic" regions plus any other resistances in series are lumped and regarded as the gate and drain series resistances. The voltages V_{GS} , V_{DG} , and V_{BS} are applied to the gate, drain, and substrate, respectively. Vertical dashed lines reflect boundaries employed in the simulation.

a floating substrate. The resistances R_s and R_D represent parasitic resistances at the source and drain and include the contact resistance and any resistance from the finite n^+ doping. These resistances can have significant effects on the MOSFET I-V characteristics when the feature size of the device is smaller than $1\text{ }\mu\text{m}$. The gate and drain voltages employed, as seen by the intrinsic transistor, are the applied gate and drain voltages less the appropriate ir drops.

The discretization of the device space into a system of grid points will be discussed in Section II.E. Based on the grid system, we shall first discuss the discretized state equations for a typical unit cell, then a new treatment of the oxide-semiconductor interface, and finally a new technique to provide high-resolution calculation without increasing the matrix size.

B. Discretized State Equations

The two-dimensional MOSFET is depicted in its simplest form in Fig. 1. A typical five-point neighborhood is shown in Fig. 2. The two-dimensional notation, such as (k,l) to denote a coordinate position (x_k, y_l) , can be conveniently replaced by a single index, so that the neighborhood of a point K consists of

$$K = (k,l),$$

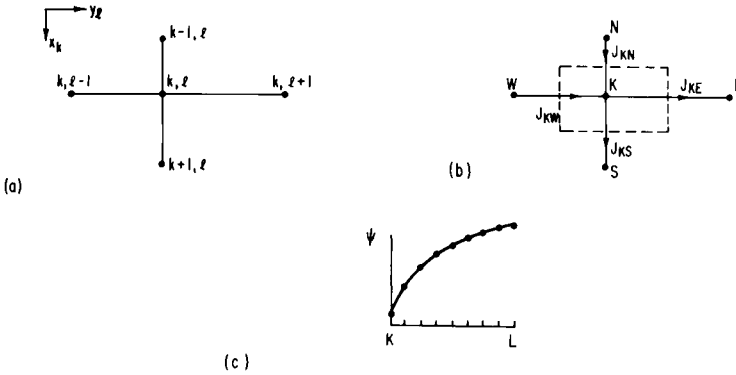


Fig. 2. (a) Two-dimensional indexing of a typical five-point neighborhood. (b) Single indexing of the same five-point neighborhood. The dashed bisectors form a rectangular unit cell for divergence of flux calculation around the point K . (c) Subdivision of the distance between the lattice points K and L into M segments for determining $(M - 1)$ high-resolution values of ψ from the two end values. The point L can be the northern, eastern, southern, or western nearest neighbor.

where L is described as

$$N = (k - 1, l), \quad E = (k, l + 1), \quad S = (k + 1, l), \quad W = (k, l - 1),$$

e.g., where the neighborhood is the point K and its northern, eastern, southern, and western nearest neighbors and where L is a generalized neighbor. This grid system is referred to as the K grid. The state variables ψ , n , and p become ψ_K , n_K , and p_K for all points K . We extend the Gummel-Scharfetter finite difference technique [40,41] to the two-dimensional case. While the state variables are solved at the grid points, the value of the current density used in the continuity equation is calculated midway between the grid points; $J = J_{K,L}$, where $L = N, E, S$, or W . The distance between K and any one of its neighbors can be different from the distance to another neighbor in order to cope with the nonuniform spatial resolution requirement of the grid (Section II.E). The state equations (5) are sparse matrices of the five-neighbor interactions, except at a grid-line termination, as discussed in Sections II.E and II.F. At the termination, the Poisson equation becomes a six-neighbor interaction and the continuity becomes a seven-neighbor interaction. However, the equations can be solved conveniently by various successive overrelaxation techniques.

Note that the Poisson matrix is asymmetric because of the unequal grid spacings and that the continuity matrix is asymmetric because of the device's directivity. The Poisson equation is iteratively solved holding n_K and p_K constant to obtain a ψ_K distribution. Then the continuity equation is solved for a new distribution n_K holding ψ_K constant. The p_K distribution, using the condition $\nabla\Phi^p = 0$ in Eq. (5), results in

$$p_K = p_K^0 \exp(-\psi_K + V_{BS} + \psi_{BI}), \quad (6a)$$

where V_{BS} is the substrate bias and ψ_{BI} the built-in voltage. The Poisson continuity iterations are repeated until a convergence condition is reached. If at an iterative stage the state variables are a set of ψ'_K , n'_K , p'_K , then we wish to find a new set $\psi_K = \psi'_K + \delta_K$ such that ψ_K satisfies the Poisson equation. Therefore, δ_K is given by

$$\begin{aligned} & \frac{2}{h_N + h_S} \left[\frac{1}{h_N} \delta_N - \left(\frac{1}{h_N} + \frac{1}{h_S} \right) \delta_K + \frac{1}{h_S} \delta_S \right] \\ & + \frac{2}{h_E + h_W} \left[\frac{1}{h_W} \delta_W - \left(\frac{1}{h_W} + \frac{1}{h_E} \right) \delta_K + \frac{1}{h_E} \delta_E \right] - a_K^2 \delta_K = b_K, \end{aligned} \quad (7)$$

where $h_L = h_{K,L}$, the distance between points K and L ,

$$a^2 = n + p, \quad b = \nabla^2 \psi' + n - p - n^0 + p^0,$$

and the discretized representation of ∇^2 is evident from Eq. (7). In

deriving (7), the approximation $e^{\psi'+\delta} = e^{\psi'}(1 + \delta)$ is used. This approximation is accurate as the iteration converges. During an intermediate or initial stage, this approximation may be gross; but it does not affect the final solution because of the properties of the Poisson matrix.

When point K is at a grid-line termination (see Sections II.E and II.F) the southern neighbor is missing. A virtual southern point is created as the average of the existing southwestern and southeastern neighbors, so that h_S and δ_S are defined.

The electron flux density in the direction from K to L can be written [from Eq. (5)] as

$$J_{K,L} = A_{K,L}(\Phi_L - \Phi_K)e^{\psi_K}, \quad (8)$$

where

$$\begin{aligned} A_{K,L} &= \left[\int_K^L \alpha \exp(-\psi_M + \psi_K) dM \right]^{-1} \\ &= \left[\alpha_{K,L} h_{K,L} \frac{\exp(\beta/2) \sinh(\beta/2)}{(\beta/2)} \right]^{-1}, \end{aligned} \quad (9)$$

$\alpha_{K,L}$ is evaluated at the midpoint between K and L , $h_{K,L}$ is the distance between K and L , and a $\beta = \psi_L - \psi_K$.

With this notation, the $\nabla \cdot J = 0$ in Eq. (5), after applying Gauss's theorem, becomes

$$\begin{aligned} n_K &\left[\frac{h_N + h_S}{2} (A_{K,E} + A_{K,W}) + \frac{h_E + h_W}{2} (A_{K,N} + A_{K,S}) \right] \\ &- \frac{h_N + h_S}{2} [A_{K,E} n_E e^{\Delta_{K,E}} + A_{K,W} n_W e^{\Delta_{K,W}}] \\ &- \frac{h_E + h_W}{2} [A_{K,N} n_N e^{\Delta_{K,N}} + A_{K,S} n_S e^{\Delta_{K,S}}] = 0. \end{aligned} \quad (10)$$

An iterative solution to n_K satisfying Eq. (10) is carried out in a manner similar to that used for ψ_K . Let $n_K = n'_K + \nu_K$, where n'_K is the value from a previous iteration. Operating on this equation with the five-point operators in Eq. (10) results in an equation for ν_K that is then iterated toward a small limit.

At a grid-line termination, the continuity equation induces a seven-point interaction. This will be treated in Section II.F.

C. The Oxide-Semiconductor Interface

At the oxide-semiconductor interface, the normal component of the displacement vector is continuous. Rather than solving for the potentials

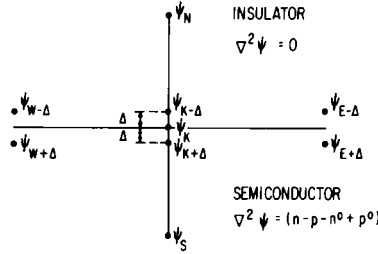


Fig. 3. The potential and position assignment used in the interface matching for the $\nabla^2\psi$ calculation.

in the two regions first and then matching the displacement calculated from these potentials, one can derive a new equation for ψ that satisfies the interface condition.

Take a small increment Δ as shown in Fig. 3 and write the discretized Laplace equation in the oxide and the Poisson equation in the semiconductor. The two equations are combined using the relation

$$\epsilon_{ox}(\psi_K - \psi_{K-\Delta})/\Delta = \epsilon_{si}(\psi_{K+\Delta} - \psi_K)/\Delta.$$

At the limit $\Delta \rightarrow 0$ an equation for δ_K similar to Eq. (7) is obtained, except that h_N , a^2 , and $(b - \nabla^2\psi')$ are replaced by $c_1 h_N$, $c_2 a^2$, and $c_2(b - \nabla^2\psi')$, respectively, where $c_1 = \epsilon_{ox}/\epsilon_{si}$ is the ratio of oxide to semiconductor dielectric constant and $c_2 = h_N/(c_1 h_N + h_S)$.

Note that the ϵ_{ox} enters the calculation only for the interfacial unit cells and not for the cells in the oxide layer. In the oxide layer, all the h s are modified by c_1 , and since a and b in Eq. (7) are both zero (Laplacian ψ), the modification factor becomes unnecessary.

D. High-Resolution Calculation

In regions where the doping gradient is large, a finer resolution than that of the K grid may become necessary for calculation of the electrostatic potential. This is because in such regions the space charge and diffusion are the dominant terms. Note that the current density is constant from K to L in the K grid; therefore, ψ may be treated as one dimensional within the KL interval. The line segment KL can be divided into M equal segments (see Fig. 2c) at points $m = 0, 1, \dots, M$. Equations (8) and (9) are valid for any subsegment. Replacing K and L by $m-1$ and m ,

$$A_{m-1,m} = [\alpha_{m-1,m} h_{m-1,m} e^\theta \sinh \theta / \theta]^{-1} \quad (11)$$

where

$$\theta = \frac{1}{2}(\psi_m - \psi_{m-1}).$$

The more accurate $A_{K,L}$ ($=A_{O,M}$) used in Eq. (10) is given by

$$[A_{K,L}]^{-1} = \sum_{m=1}^M (A_{m-1,m})^{-1} \quad (12)$$

for any direction KL .

To obtain the one-dimensional ψ_m , the values of n_m and n_m^0 are needed. The latter can be read from the device description input file or adequately approximated by

$$n_m^0 = n_0^0 \exp[(m/M) \ln(n_0^0/n_0^0)]$$

for $m = 1, 2, \dots, M-1$; $n_0^0 = n_K^0$, and $n_M^0 = n_L^0$.

The values of n_m are obtained from the one-dimensional continuity equation,

$$n_m = \left\{ A_{K,L} \sum_{m=1}^m (A_{m-1,m})^{-1} [n_L \exp[-(\psi_L - \psi_K)] - n_K] + n_K \right\} \exp(\psi_M - \psi_K) \quad (13)$$

for $m = 1, 2, \dots, M-1$; $n_0 = n_K$, $n_M = n_L$.

The one-dimensional Poisson equation relating the ψ_m and the space charge $n_m - n_m^0$ is then solved via the forward-elimination and backward-substitution technique:

$$\begin{aligned} s_m &= m/(1+m), \\ d_m &= n_m^0 - n_m \quad (\text{except } d_1 = n_1^0 - n_1 + \psi_1), \\ e_m &= s_m(d_m + e_{m-1}) \quad (e_0 = 0) \end{aligned} \quad (14)$$

for $m = 1, 2, \dots, m-1$. Thus

$$\psi_m = s_m \psi_{m+1} + e_m$$

for $m = M-1, M-2, \dots, 1$.

In using Eqs. (12) and (13), we solve for n and ψ with M times the spatial resolution of the main grid size. However, the major matrices with K, L indices do not increase in size. This is in contrast with Slotboom's [35] scheme of increasing the size of the Poisson matrix to obtain higher resolution for ψ .

E. Grid Representation of MOSFET Devices

The two-dimensional MOSFET is depicted in its simplest form in Fig. 1. The computer program is capable of simulating reasonable modifications of this structure, such as unequal depths of source and drain doping, a graded $p-n$ junction, and arbitrary doping distribution in the bulk. Using the pictorial model of Fig. 1, we note that in the region of interest,

i.e., the channel region, the ψ spatial resolution must be very fine to describe the rapidly varying carrier density, which may vary by several decades per unit voltage variation in the electrostatic potential, which in turn can have a steep spatial gradient. Thus a spatial resolution on the order of 10^{-6} cm is called for in simulation of the MOSFET of the $1\text{-}\mu\text{m}$ range of channel length. In order to simulate two-dimensional properties, it is necessary that the y spatial resolution be of the same order of magnitude or else the great disparity in resultant matrix values would lead to poor accuracy and a poor convergence rate (in the relaxation solution). On the other hand, the actual device dimensions are of the order of microns, and a uniform spatial resolution of 10^{-6} cm would require a very large number of grid points, making the simulation costly and impractical. Thus the grid size must be variable.

Study of the problem and simulation experience has led us to the grid structure shown in Fig. 4. The following rules are generally, but not always, observed.

(1) The device width Y is divided into 64 equal line segments, the middle 40 of which represent the channel length. For example, a $1\text{-}\mu\text{m}$ channel-length device requires $Y = 1.6\text{ }\mu\text{m}$ and $\Delta y = 0.025\text{ }\mu\text{m}$. The

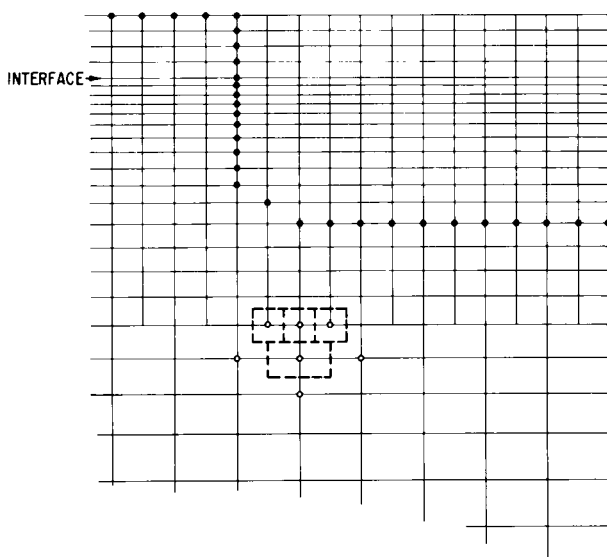


Fig. 4. The device in Fig. 1 is discretized into grid structure suitable for simulation; only the upper right corner is shown. Known potential values are assigned on each dot. The grid points outside the dotted boundary (i.e., upper right corner and a similar region at the source corner in Fig. 1) are not relevant to the simulation. The dashed boxes show the unit cells where grid-line termination plays a role, as discussed in Section II.F.

channel length is represented by more or less than 40 segments as desired.

(2) The first Δx incremental size under the oxide–semiconductor interface is 2.5 times smaller than Δy . In the above example, Δx at the interface is $0.01 \mu\text{m}$.

(3) The Δx incremental size in the semiconductor is 10% larger than that immediately above, $\Delta x_{i+1} = 1.1 \Delta x_i$.

(4) When Δx becomes larger than Δy , the latter is increased by a factor of 2. This is accomplished by terminating the appropriate vertical grid lines. The 65 initial grid lines are reduced to 33, 17, 9, 5, and 3 as required. The $\Delta x:\Delta y$ criterion is not rigid; Δx can be as large as $2.5\Delta y$ if desired.

(5) In practice, the oxide layer thickness is divided into 4 Δx_{ox} of equal size. There can be more than four layers if desired.

In this manner, a total of 40 horizontal grid lines can represent a total of $4 \mu\text{m}$ in thickness for a $1\text{-}\mu\text{m}$ channel-length device and give sufficiently high spatial resolution in the relevant regions. This grid system has proved sufficient for simulating MOSFET devices of interest.

It is further recognized, based on Figs. 1 and 4, that because of the finer resolution many grid points lie (i) in the n^+ regions of the source and drain and (ii) in the oxide layer above the source and drain n^+ regions. It is reasonable to assume that in group (i) the potential is constant and the region can be treated as metallic. There is no need to simulate the potential and carrier distributions for these regions. In group (ii) points there are no carriers, and although the potential distribution may vary in these regions, it does not significantly affect the static behavior of the device. Hence, the device region of interest can be bounded by the dots in Fig. 4. The potential along the line joining the gate edge to the drain edge is a linear function of distance. Similarly, the potential is given along the line joining the gate edge to the source edge. Hence, the Dirichlet boundary condition can be applied to the line boundaries from the gate contact to the side-wall edges of the source and drain n^+ regions; known potential values are assigned.

With such a known upper boundary and known lower boundary (given by the substrate bias plus appropriate built-in voltage), the Poisson equation can be solved for the potential distribution ψ with the additional side-wall boundary condition $\partial\psi/\partial y = 0$. (The discontinuity at the oxide–semiconductor boundary was treated in Section II.C.)

The electronic carrier distribution, assuming an n -channel device on a p -channel substrate as in Fig. 1, can be solved from the transport equation with the following boundary conditions:

(a) $n = \text{const}$ (assigned, say, $3 \times 10^{19} \text{ cm}^{-3}$) at the ohmic boundaries of the source and drain.

(b) The flux normal to the oxide–semiconductor interface is 0.

- (c) Reflecting flux boundary condition exists at the vertical side walls.
- (d) The vertical flux is continuous at the lower boundary.

Once the ψ and n distributions are known, the hole distribution can be determined using the relationship between ψ and p and the Fermi potential in the equilibrium state (no avalanche and recombination-generation), as discussed in Section II.B.

Of the $65 \times 40 = 2600$ "full" grid points, many are skipped in the variable-grid-size scheme just discussed. The numbers of grid points that need to be calculated for ψ and n are different; there is no need to solve for n above the interface line in Fig. 4.

F. Single Indexing of Two-Dimensional Grid Points

The relevant grid points at which ψ and n (and hence p) distributions are calculated are then given single-index serial numbers comprising the K grid, as described in Section II.B. There are more " ψ " points than " n " points since ψ solutions are needed in the oxide layer. A file is prepared in which each grid point is assigned the serial numbers of its northern, eastern, southern, and western neighbors, as well as a point-type index to indicate whether the grid point is an internal point or a boundary point and, if the latter, which kind of boundary. When a point is situated at the vertical line termination, there is no southern neighbor and a virtual grid point is simulated in the calculation, such as

$$2\psi''_{IS(K)} = \psi_{IS(K+1)} + \psi_{IS(K-1)},$$

where $IS(K)$ designates the serial number of the southern neighbor of grid point K . The quotation marks indicate a virtual point.

The size of the unit cell surrounding each grid point is also tabulated in the file. Figure 4 shows the unit cells used in setting up the transport (current continuity) equation. Note that for the cell below a vertical grid-line termination the fluxes from the three northern cells are weighted by $\frac{1}{4}$, $\frac{1}{2}$, and $\frac{1}{4}$ from left to right, reflecting the fractional length of the northern cell boundary.

The information on the donor and acceptor concentrations at these grid points are also tabulated so that the serial grid file contains all relevant information on each point.

Such serialization of the relevant grid points allows the simulation to be carried out in two steps. The device structure is programmed into the grid construction program that produces the serial number code file. The code file represents the complete device. A separate program is then called on

to solve the matrix equations as a function of the operating parameters, i.e., the gate, drain, and substrate voltages.

The serialized grid points are convenient for matrix inversion either by direct inversion, such as is often employed in the finite-element technique, or by an iterative technique, especially when only nearest-neighbor interactions are considered in calculating the matrix elements.

The serialized grid numbers provide the natural DO LOOP index for the relaxation technique. In the SOR technique, updated information is fed into the K position from its western and northern neighbors in the forward sweep (and hence more strongly coupled to the western and northern boundary conditions) and from its southern and eastern neighbors in the reverse sweep (with better coupling to the southern and eastern boundary conditions). Thus reversing the sweep direction after each iteration tends to reduce any rectification effect and produces faster convergence. A sweep-down, sweep-up line-relaxation technique was also tried; there was no noticeable improvement in convergence rate or accuracy. Because of its simplicity and ease in program modification, the point-relaxation technique is preferred.

The overall logistics of the MOSFET simulation is shown in Fig. 5, which indicates the optional interfacing with the well-known SUPREM program [11].

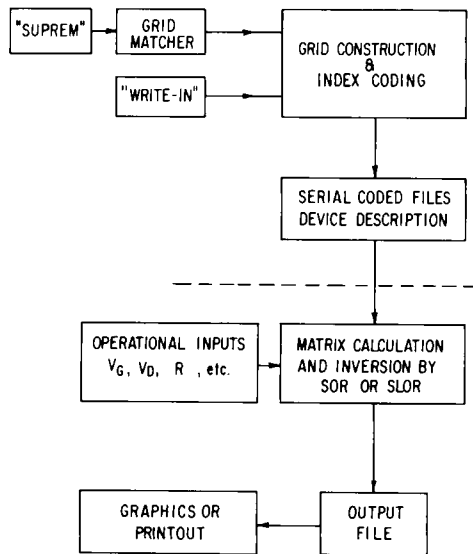


Fig. 5. The overall logistics of the MOSFET simulation.

G. Material Parameters and Calculation of the A Coefficients

Aside from the normalization factors discussed in Section II.A, the only material parameter involved in the POISSON portion of the program is the dielectric constant discontinuity at the interface. The CONTINUITY portion of the program involves material parameters more directly, namely the velocity-field characteristic and, in case of avalanche multiplication, the ionization coefficients.

The current flux density is to be evaluated at the unit cell boundaries, midway between the adjacent grid points. This is carried out in terms of the A coefficients defined in Section II.B, which include the parameter α , defined as the inverse of the carrier mobility, which, in turn, is defined as the ratio of the velocity and the electric field.

For simulation of silicon devices, the $v(E)$ characteristics of the electrons and holes are required. From the log-log plot of the $v(E)$ characteristic given by Sze [50], the curve is piecewise fitted with simple analytical functions of the type

$$v = \begin{cases} \mu_0 E^* & \text{for } E^* \leq E_c^*, \\ v_{\max} - w \exp[-(E^* - E_c)] & \text{for } E^* > E_c^*, \end{cases}$$

where $v_{\max} = 10^7$ cm/sec, $w = 7 \times 10^6$ cm/sec, $E_c^* = 2000$ V/cm, and $\mu_0 = 1.5 \times 10^3$ cm²/V sec for electrons. The asterisk denotes the effective electric field defined analytically to include empirical dependence of the mobility on the background doping level [50, p. 40] and on the electric field normal to the oxide-semiconductor interface [51]. The effective field is defined as

$$E^* = EE_1E_2,$$

where E , the magnitude of the electric field, is equal to $\partial\psi/\partial s$. For a spatial increment ∂s ,

$$E_1 = \exp\{-[0.12763 \ln(p^0 + n^0) - 5.401]^2\},$$

where p^0 and n^0 are, respectively, the background acceptor and donor concentrations in reciprocal cubic centimeters, and

$$E_2 = 1 - (1.34 \times 10^{-7})E_p + (6.242 \times 10^{-15})E_p^2,$$

where E_p is the field perpendicular to the oxide-semiconductor interface in volts per centimeter. The value of E_2 is limited to $0.5 \leq E_2 \leq 1.0$ for interface cells and is unity for all other cells.

By so writing E^* , the log-log shape of the $v(E^*)$ characteristic as well as the value of the saturation velocity are maintained. Hence the piecewise $v(E^*)$ subroutine need not be changed; only E^* renormalization is

required, as previously given. In fact, the $v(E)$ characteristic for holes can be obtained with the same subroutine by using an additional electric field renormalization factor $E_3 = \mu_0(\text{hole})/\mu_0(\text{electron}) = 0.32$.

Having determined v as a function of E via the E^* , we define the quantity α as

$$\alpha = \left| \frac{\partial \psi}{\partial s} \right| / v,$$

and it is determined at the midpoint along ∂s from the potentials at the ends of the ∂s increment. Thus the A coefficients can be calculated for the northern, eastern, southern, and western directions of the flux.

H. Implementing the High-Resolution Calculation

The A coefficient for the flux in a given direction is a function only of the potential difference spanning the length increment, and the electric field value used in the α determination is taken as $\Delta\psi/\Delta s$. The accuracy decreases for large $\Delta\psi$ because the carrier density depends exponentially on ψ . In Section II.D, a solution was discussed for n and ψ distributions that is internally consistent along an axis *within* a unit cell. The unit cell is subdivided into M subsections along the L direction ($L = \text{N, E, S, or W}$) and along each direction a one-dimensional boundary-value problem is assumed and is completely determined. Since α is determined within the m th subsection using the determined $\Delta\psi_{m-1,m}$, the $A_{m-1,m}$ is determined so that for the cell K

$$\frac{1}{A_{K,L}} = \sum_{m=1}^M \frac{1}{A_{m-1,m}},$$

where L may represent the N, E, S, or W direction. Note that M can be different for different L .

The need for more accurate calculation of the A coefficients arises when the Debye length of the carrier is much smaller than the unit cell dimension. This occurs near the ohmic source and drain regions, where the doping density is high and the magnitude of the density gradient is large. The choice of the approximate value of M , therefore, is a function of doping as well as of the gradient. In the channel region, it is adequate to choose $M = 1$ (no subdivision) if $n^0 \leq p^0$, where p is the substrate doping, and to choose $M \approx \ln[(n_K^0)^2 - (n_L^0)^2]$ for larger n^0 values. The M is kept less than 10 because larger values do not improve the accuracy appreciably. During the iteration, the initial n and ψ distributions may be so inaccurate that within a unit cell the one-dimensional subdivision solution

becomes unstable. This is prevented by placing a limit on the magnitude of the d matrix in Equation (14):

$$d_m \leq (4/M)|\Delta\psi_{K,L}|/(h_{K,L})^2.$$

III. RESULTS AND DISCUSSION

The MOSFET is relatively well understood in its qualitative characteristics. When two-dimensional effects are significant, as in a short-channel MOSFET, quantitative characteristics are hard to calculate and computer simulation is needed. The accuracy of our computer simulation will be checked by comparison with experimental results in Section III.A, and the results will prove to be excellent. A study of the short-channel effects via simulation will be presented in Section III.B, and the parasitic resistance effect will be discussed in Section III.C.

A. Comparison with Experimental Results

The model has been tested in some details against published results in Dennard, Gaensslen, Walker, and Cook [52]. The substrate acceptor doping profile believed to be close to that in the real n -channel device is shown as profile A in Fig. 6, and the gate length is taken to be $1.3\ \mu\text{m}$. Comparison of the I - V characteristic is shown in Fig. 7. Because avalanche generation has not been included in the program, the simulation was not carried beyond $V_{\text{DS}} = 5\ \text{V}$. The agreement is quantitatively very good. Also shown in Fig. 7 is the I - V characteristic of a p -channel device of the same structure for $V_{\text{GS}} = 3\ \text{V}$. The current is lower than that in the n -channel device for the same gate voltage because hole mobility is lower than electron mobility.

The threshold properties of this n -channel device were also investigated. Figure 8 compares the drain current calculated as a function of the gate voltage at a low drain voltage with the experimental drain current. The threshold voltage of $0.68\ \text{V}$ and the slight curvature at $V_{\text{GS}} \approx 0.9\ \text{V}$ are reproduced by the simulation. The subthreshold current for this device is shown in Fig. 9, where again the quantitative agreement between simulated and experimental values is excellent. The semilogarithmic slope is $100\ \text{mV/decade}$. Comparison with other short-channel devices shows similar agreement [46].

The equipotential and equicarrier density contours in the device under $V_{\text{GS}} = 3\ \text{V}$, $V_{\text{DS}} = 5\ \text{V}$ condition are shown in Fig. 10. The field reversal position under the gate in Fig. 10a and the two-dimensional depletion

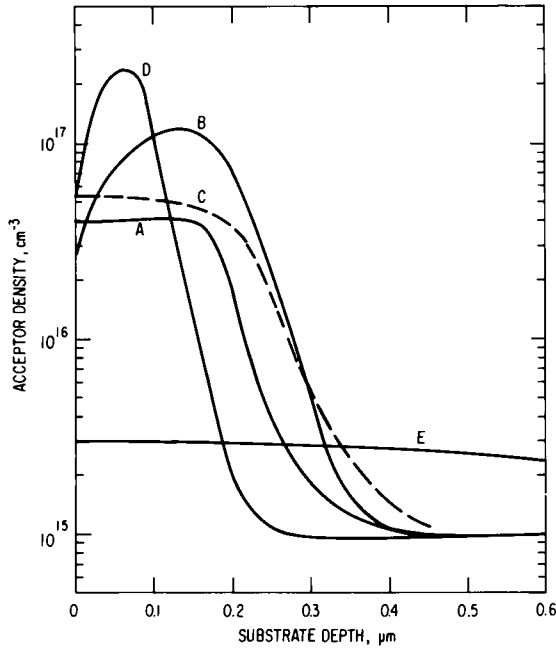


Fig. 6. Various substrate doping profiles used in simulation.

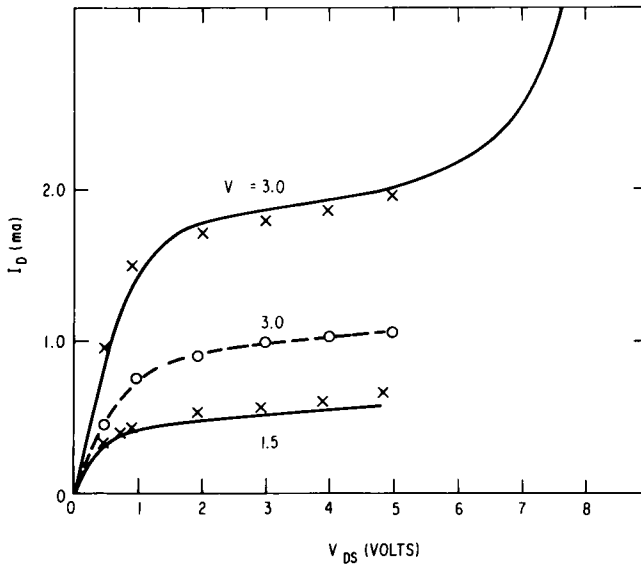


Fig. 7. Simulated (×) and experimental (—) transfer characteristics of an *n*-channel MOSFET with 1.3- μm channel length. The dashed curve represents the simulation result for a *p*-channel device of the same construction showing low current due to lower hole mobility. The values used were $L = 1.3 \mu\text{m}$, $W = 15.3 \mu\text{m}$, and $d_{ox} = 250 \text{ \AA}$.

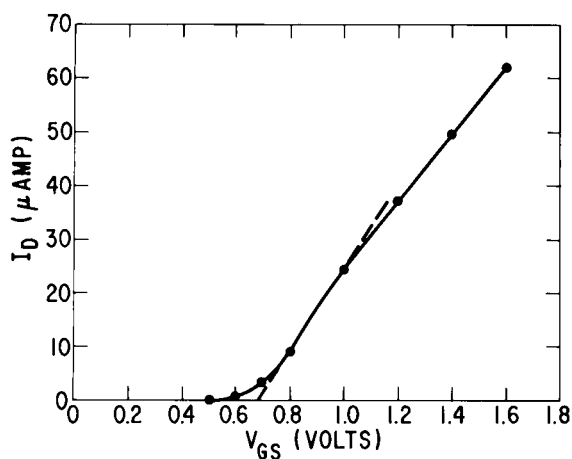


Fig. 8. Comparison of the simulated (●) and experimental (—) drain current at low drain voltage (0.1 V) as a function of the gate voltage for the 1.3- μm n -channel device, with $W = 15 \mu\text{m}$ and $d_{ox} = 250 \text{ \AA}$. The dashed line is used for extrapolating to the threshold voltage.

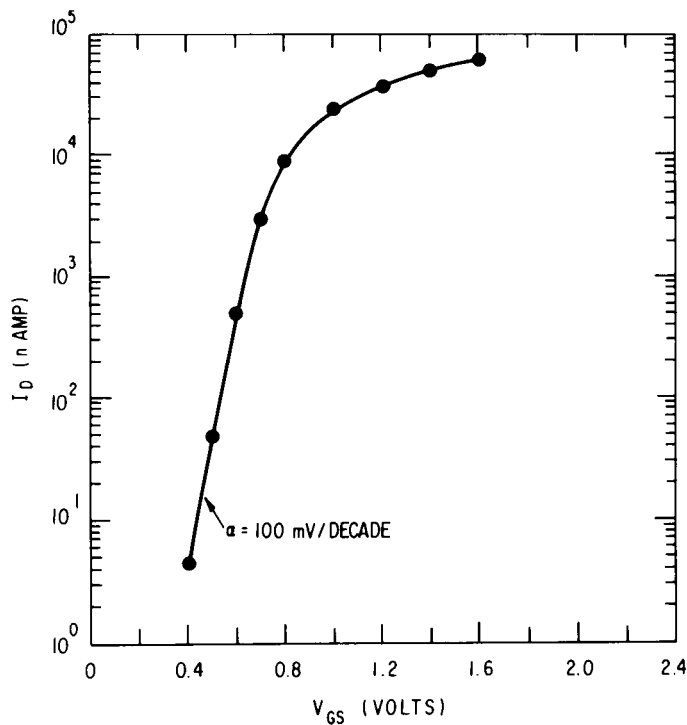


Fig. 9. Simulated subthreshold current as an exponential function of the gate voltage for the 1.3- μm n -channel device at low drain voltage (0.1 V) and $W = 15 \mu\text{m}$ and $d_{ox} = 250 \text{ \AA}$. The semilogarithmic slope of 100 mV/decade agrees well with experimental value.

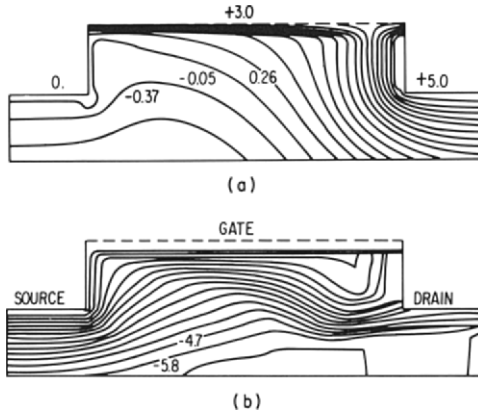


Fig. 10. (a) The equipotential contours (0.315 V/step); (b) The equicarrier density contours (1.05 decades/step) in the 1.3- μm n -channel device. Intrinsic level corresponds to 0.

region in Fig. 10b illustrate the two-dimensionality aspects of the simulation.

Based on quantitative agreements with the experimental results, it is concluded that the model is quantitatively accurate and, therefore, can be used for design and diagnostic purposes.

B. Short-Channel Effects

Having established the accuracy of our model, we use it to study the challenging problems in device designing and circuit modeling in VLSI. The relevant parameters are the threshold voltage and I - V characteristics of the device, which depend on both the doping distribution and the operating voltages. The effect of parasitic resistance also becomes more pronounced in short-channel devices. To illustrate short-channel effects, we compare the performance characteristics of devices of the same doping design (profile B in Fig. 6) for 0.5 and 3- μm channel lengths. The 0.5- μm value is chosen because it is usually considered as the lower limit for practical Si MOSFETs [25,53] and the 3- μm value can be considered as a long device. As shown in Fig. 11, when the channel length is small, the source-drain current in the saturation region is no longer completely controlled by the gate voltage. The increase in the drain current with drain voltage in the short-channel device may be interpreted as a change in the threshold voltage caused by the drain voltage [45]. The effect should depend on the substrate doping profile.

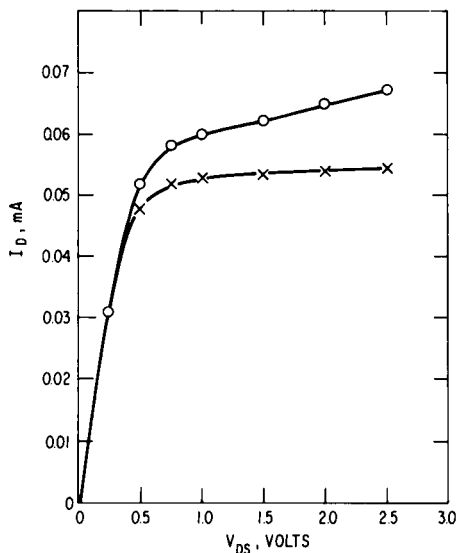


Fig. 11. Drain current versus drain voltage for a long gate ($\times$, $L = 3 \mu m$) and a short gate ($\circ$, $L = 0.5 \mu m$) device built on the substrate with profile B of Fig. 6. The oxide thickness is 2×10^{-6} cm. Source and drain junctions are 2×10^{-5} cm below the interface. The gate voltage is 2 V and the current is scaled for channel width equal to channel length.

To study the sensitivity/immunity of devices with different doping profiles to the short-channel effects, doping profiles B and C in Fig. 6 were used. The threshold voltages of the devices of 0.5- and 3- μm channels with profiles B and C were investigated at two drain voltages. It might be expected that the device with doping profile B would be less sensitive to the short-channel effects than that with doping profile C.

The threshold voltage sensitivity to the effect of shorter channels is illustrated in Fig. 12, where I_D versus V_{GS} characteristics at low V_{DS} (0.1 V) are shown at the two channels for the two doping profiles. The threshold voltage for the profile-C device shifts from 0.575 to 0.524 V when the channel is shortened from 3 to 0.5 μm , while the shift is only from 0.65 to 0.63 V for the profile-B device.

The threshold shift in the lower doping (profile C) device is much more influenced by the drain voltage than in the higher doping (profile-B) device, as shown in Fig. 13. Here, with $V_{DS} = 3$ V, the short-channel effect is very marked in the profile-C device; the threshold shifts from 0.53 V for the 3- μm gate length to 0.255 V for the 0.5- μm gate. The profile-B device is less susceptible to the short-channel effect; the threshold shifts only from 0.55 to 0.49 V. In addition to the short-channel effect, Fig. 13 also

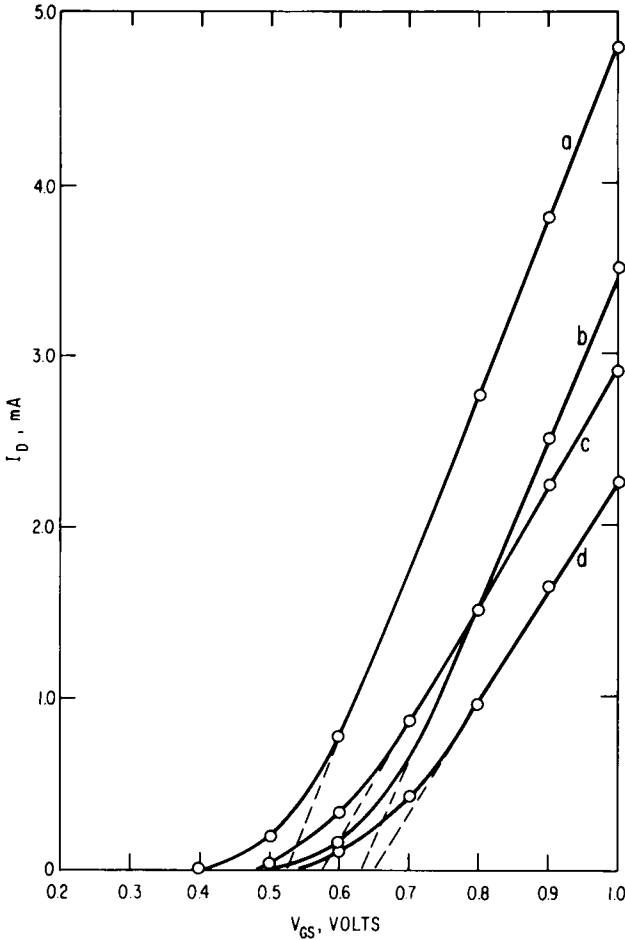


Fig. 12. Drain current at low drain voltage (0.1 V) versus the gate voltage for short- and long-channel devices: (a) 0.5- μm channel, profile C; (b) 3- μm channel, profile C; (c) 0.5- μm channel, profile B; (d) 3- μm channel, profile B. Note that the calculated points form smooth curves.

shows that square law dependence of drain current on gate voltage in the saturation region holds for short-channel devices.

Equipotential contours and hole equidensity contours for the short-channel devices studied in Figs. 12 and 13 are given for the two drain voltages at a single gate voltage of 0.4 V in Figs. 14 and 15. The effects of lower doping profile and high drain voltage are negligible in the long gate device, as shown in Fig. 16. The hole carrier density (logarithmic scale)

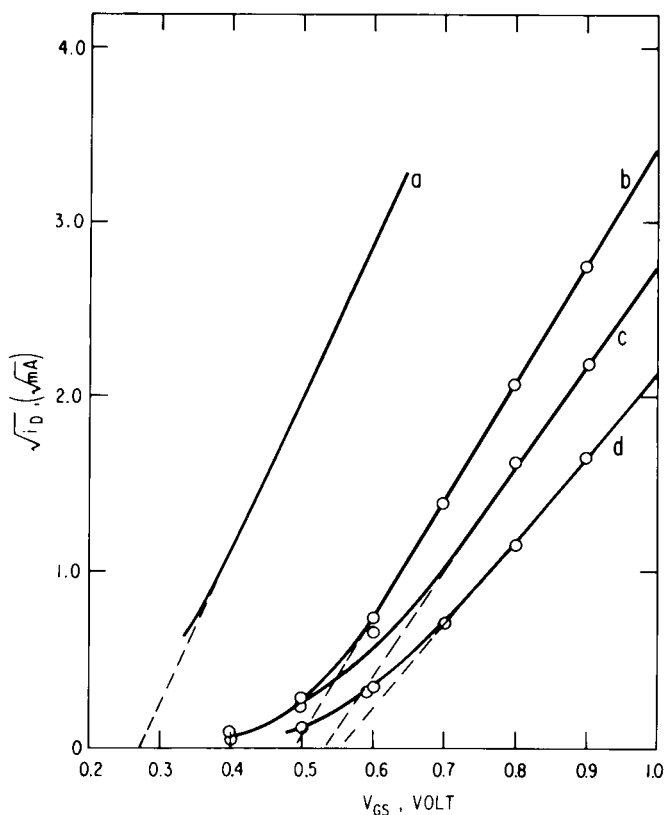


Fig. 13. Square root of the drain current at high drain voltage (3 V) versus the gate voltage for short- and long-channel devices (notations as in Fig. 12). Note that the square root of the drain current is used under the current saturation cases $V_{GS} < V_{DS}$.

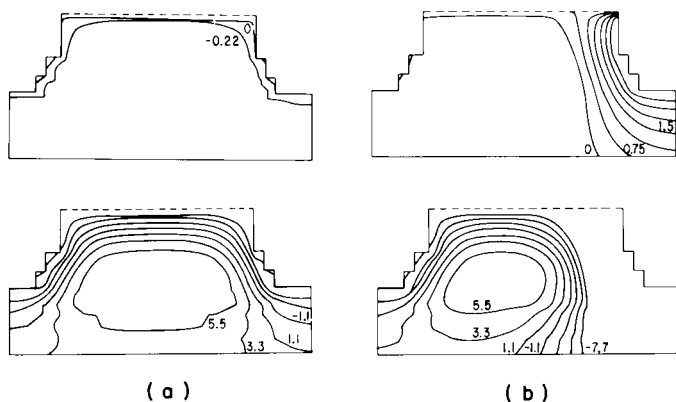


Fig. 14. The equipotential contours (top, linear steps) and equicarrier density contour (bottom, logarithmic steps) in the devices with profile B (in Fig. 6), 0.5- μm channel, and $V_{GS} = 0.4$ V: (a) $V_{DS} = 0.1$ V; (b) $V_{DS} = 3$ V. Only the portion of the device near the channel is shown. Not all contours calculated are plotted.

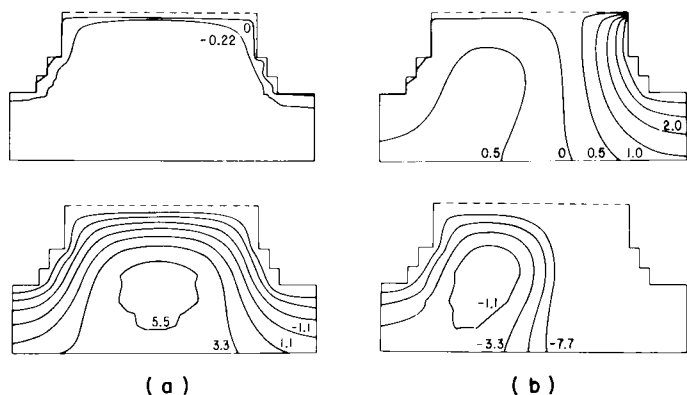


Fig. 15. Similar plots as in Fig. 14 for devices with profile C in Fig. 6.

contours are depicted instead of the electronic contours to better illustrate the extent of the space-charge regions. The short-channel effect is qualitatively evident in these figures as is the degree of spreading of the space-charge layers from the source and drain electrodes; the threshold shift is a manifestation of these changes.

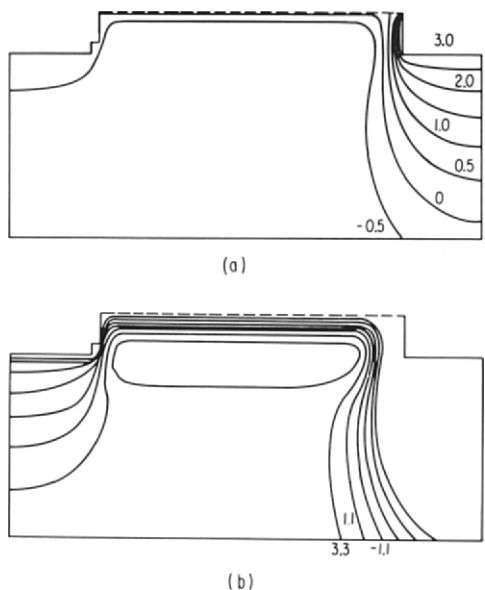


Fig. 16. Equipotential (a) and hole equidensity (b) contours in a 3-μm channel device of profile C, with $V_{GS} = 0.4$ V and $V_{DS} = 3$ V.

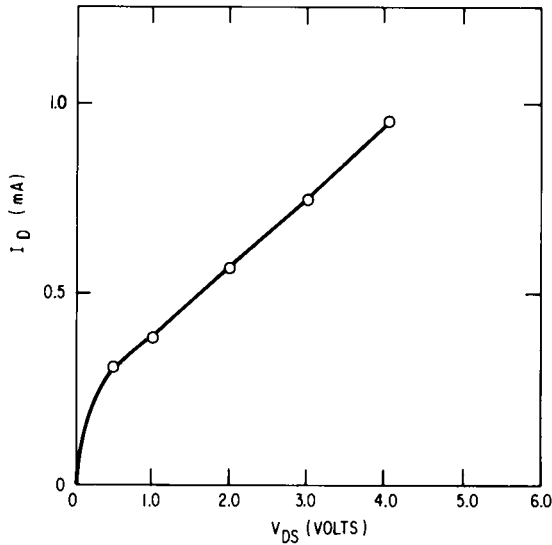


Fig. 17. Drain current versus drain voltage of device with profile E at 0.4-V gate voltage and zero substrate voltage.

The merit of the simulation has been demonstrated in a practical design problem. A design was in progress for a chip based on a $1.45\text{-}\mu\text{m}$ MOSFET with $400\text{-}\text{\AA}$ oxide thickness on a substrate with doping profile E in Fig. 6. Our simulation results for this MOSFET are shown in Figs. 17–20. The simulated threshold voltage and the I – V characteristics are

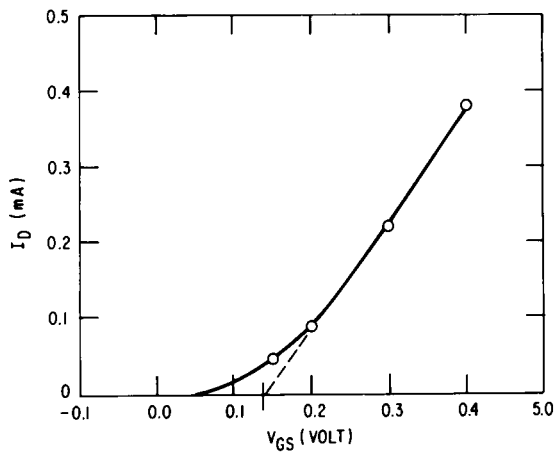


Fig. 18. Drain current versus gate voltage of device with profile E at 1-V drain voltage and zero substrate voltage.

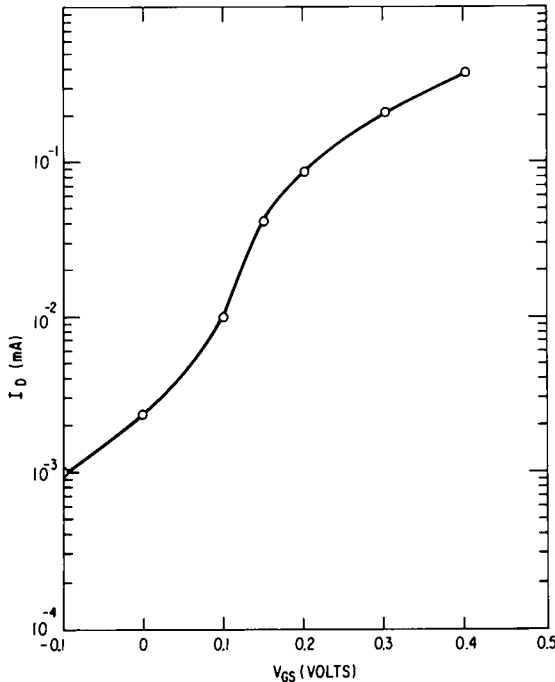


Fig. 19. Logarithmic plot of drain current versus gate voltage in the subthreshold range showing poor threshold characteristic of device with profile E at 1-V drain voltage and zero substrate voltage.

related to the very extensive space-charge regions of the source and drain, as depicted in Fig. 20. Because the substrate doping is too low, the short-channel effect in this 1.45- μm channel device is severe; its characteristics are known as those of a punched-through device.

C. Parasitic Resistance Effect

For devices with feature size below 1 μm , the contact resistances at the source and drain may become significant compared with the channel resistance. Using a specific contact resistance of $10^{-5} \Omega \text{ cm}^2$ and assuming that the contact windows for the source and drain are the same size as that of the channel, the contact resistance would be 0.2 and 0.4 $\Omega \text{ cm}^2$ (per centimeter channel width), respectively, for 0.5 and 0.25- μm channel devices.

The effect of the parasitic or contact resistances is illustrated in Fig. 21 for such devices. The substrate doping profiles are given as profiles B and

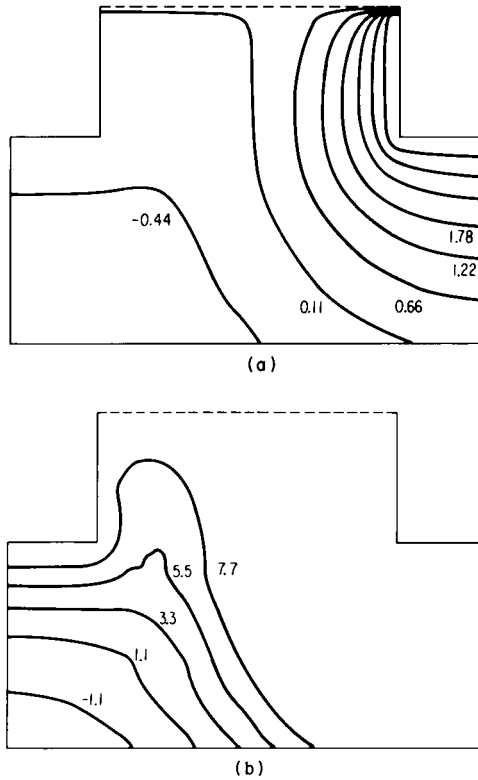


Fig. 20. (a) Equipotential contours in device with profile E at $V_{DS} = 4$ V, $V_{GS} = 0.4$ V, and $V_{BS} = 0$. (b) Equihole density contour (logarithmic scale) in device with profile E at the same operating condition as in (a).

D in Fig. 6. The device dimensions, oxide thickness, and doping profile are scaled according to the constant electric field rule.

In Fig. 21, I_{DS} versus V_{DS} characteristics are given for the 0.5 (device B) and the 0.25- μm devices (device D), with and without the parasitic resistances. Without the parasitic resistance, the drain-current ratio is approximately 2, in agreement with that expected from the scaling rule. ($I_{DS} \approx 62 \mu\text{A}$ at 2 V for device B and $28 \mu\text{A}$ at 1 V for device D.) However, when the parasitic resistances are accounted for, the drain currents decrease, respectively, to 79 and 50% for devices B and D, and the specific drain-current ratio is 3.68.

The performance degradation by the parasitic resistances in submicrometer channel devices is mainly caused by the debiasing effect of the voltage drop across the source resistance. The constant electric field scaling rule cannot conveniently take this into account and hence cannot be used

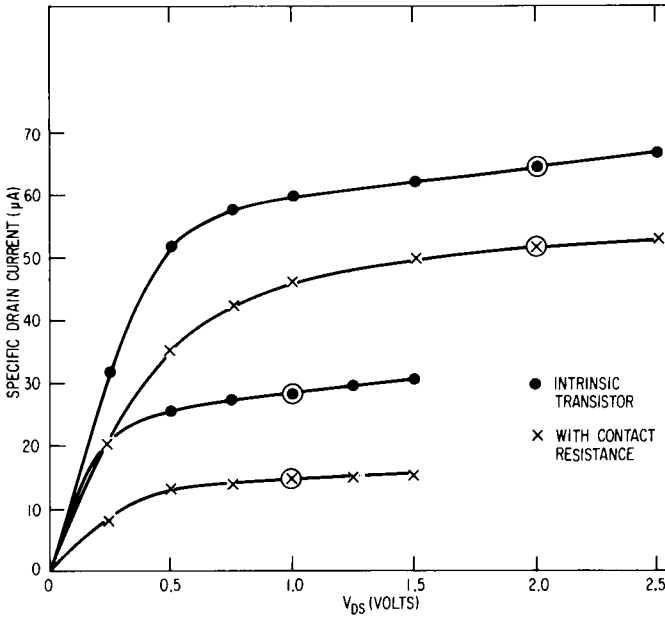


Fig. 21. The transfer characteristics of profile B and D devices as a function of gate length and parasitic resistance. Upper pair is a $0.5\text{-}\mu\text{m}$ gate, $V_{GS} = 2\text{ V}$; lower pair is a $0.25\text{-}\mu\text{m}$ gate, $V_{GS} = 1\text{ V}$. The circled points are used in the specific drain-current ratio calculation.

to predict the performance of submicrometer devices. More importantly, the results shown here, as well as other considerations [25,53], indicate that one reaps a diminishing return in reducing the Si MOSFET channel length below $0.5\text{ }\mu\text{m}$. With present MOSFET physical configuration, the parasitics (resistances and capacitances) overwhelm the active device in this region.

IV. CONCLUSIONS

Quantitatively accurate computer modeling of the short-channel MOSFET, the key element in VLSI, is now available. The excellent agreement between the simulated and experimental device performances testifies to the correctness of the model and the material parameters used. The performance of the active device under a design rule of $3\text{ }\mu\text{m}$ or less (down to $0.5\text{ }\mu\text{m}$) can be simulated, including the effect of the parasitic resistances. Device design variables, such as the doping profile, junction

depth, and gradient, can be tested in the simulation. Great savings in time, design, and processing costs should accrue.

There are several attributes that make possible the accuracy of the simulation at a reasonable computing cost. Most notable is the complementary natures of the mathematical formulation and the grid system employed. The latter provides a very high spatial resolution in the critical regions while suppressing the total number of points. It also facilitates higher speed in the matrix inversion by maintaining the aspect ratio of all unit cells near unity and by allowing the subdivision of space within the unit cell to achieve ultrahigh resolution (approaching the local Debye length) when necessary, without increasing the matrix size. The latter is in contrast to the prevailing practice, where higher grid density is required. The novel treatment of the oxide-semiconductor interface also has contributed to the improved accuracy.

The speed of the iterative convergence of the present program compares favorably against other reported programs [35,39]. Savings in computing cost can be realized in cases in which the degree of accuracy may be less, such as in test runs for qualitative behaviors and/or "ball-park" results. For high accuracy, a converged result requires approximately 10 min of a Honeywell 6000 CPU time.

The simulation results obtained demonstrate the necessity of the two-dimensional considerations. The cost of running an accurate two-dimensional program such as ours is minimal and, in fact, may represent a great savings in time and other costs compared with the cost of a possible failure in a device design without the simulation.

Future extension of the simulation capabilities includes the SOS MOSFET boundary conditions and additional device physics. Inclusion of the two-carrier transport resulting from avalanche multiplication, recombination-generation, trapping, and hot-carrier effects is under consideration. Modification of the grid structure is also possible in order to apply the program to other devices.

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